

Parallel lines

Two lines in a plane that never meet — and how you can tell without following them.

LESSON 81

1 × 40 MIN

MATEMATIKA 7, §37

S8 5.1

LESSON CLOCK

8'

12'

12'

6'

2'

The definition	8 min
The axiom of parallels	12 min
Always the same distance	12 min
Around the room	6 min
Homework	2 min

LEARNING OBJECTIVES

- Define parallel lines and use the symbol $\parallel$.
- Distinguish parallel lines from lines that merely look apart.
- State the axiom of parallels and what it guarantees.
- Recognise parallel lines in the figures and objects around the room.

TEXTBOOK

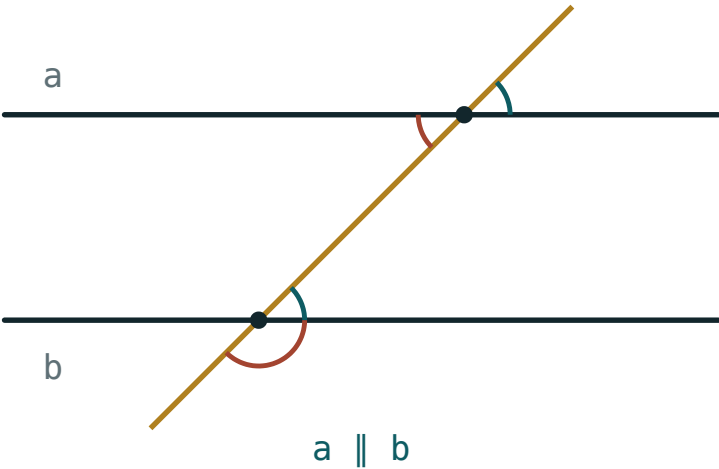
National	pp. 226–230
Cambridge	Stage 8, pp. 44–47
Workbook	Exercise 5.1

01

Explanation

The definition

Two lines in the same plane are **parallel** if they have no common point, however far they are produced in either direction. We write $a \parallel b$.



Two parallel lines and a transversal

TWO LINES IN A PLANE	COMMON POINTS	NAME
they cross	1	intersecting
they never meet	0	parallel
every point is shared	all	the same line

“THEY DO NOT MEET ON MY PAGE” PROVES NOTHING

A line has no end. Two lines drawn a little apart may cross far outside the page, so parallelism can never be checked by looking — it has to be established from angles, which is the next lesson.

The axiom of parallels

Through a point not on a line there passes exactly one line parallel to it.

“At least one” can be proved; “exactly one” cannot, and is taken as an **axiom**. It is the most famous statement in geometry — mathematicians tried for two thousand years to derive it, and the failure produced the non-Euclidean geometries.

CLAIM	STATUS
at least one parallel exists	a theorem
there is no more than one	the axiom
parallels stay the same distance apart	follows from the axiom

WHY THE AXIOM IS NEEDED

Without it a triangle's angles need not add to 180° . Every result of the next three lessons rests on it, which is why it is stated openly rather than assumed silently.

Always the same distance

The distance from a point to a line is measured along the perpendicular. If $a \parallel b$, that distance is the same wherever the point is taken on a .

LINES	DISTANCE BETWEEN THEM
parallel	constant
intersecting	falls to zero at the crossing

THIS IS HOW RAILS AND RULED PAPER WORK

A railway is built to a constant gauge precisely so that the rails are parallel; if the distance changed anywhere, the wheels would jam or drop. Constant distance and parallelism are the same fact stated two ways.

Around the room

OBJECT	PARALLEL PAIR	WHY IT MATTERS
ruled paper	the lines	writing stays level
a railway	the rails	constant gauge
a ladder	the two side rails	the rungs stay square
a window frame	opposite edges	the pane fits
a book's opposite edges	top and bottom	the pages stack

In each case the parallelism is not decoration: it is what makes the object work.

02 Worked examples

EXAMPLE 1 How many lines through a point M outside a line a are parallel to a ?

- ① At least one exists — this can be constructed.
- ② The axiom says there is no second.
- ③ So the number is exactly one.

ANSWER

1

EXAMPLE 2 Two lines in a plane are drawn 3 cm apart at one point and 3.4 cm apart at another. Are they parallel?

- ① Parallel lines keep a constant distance.
- ② Here the distance changes.
- ③ So they are not parallel — they meet somewhere.

ANSWER

No

EXAMPLE 3 If $a \parallel b$ and $b \parallel c$, what can be said about a and c ?

- ① Suppose a met c at a point P .
- ② Then through P there would be two lines parallel to b .
- ③ That contradicts the axiom.
- ④ So $a \parallel c$.

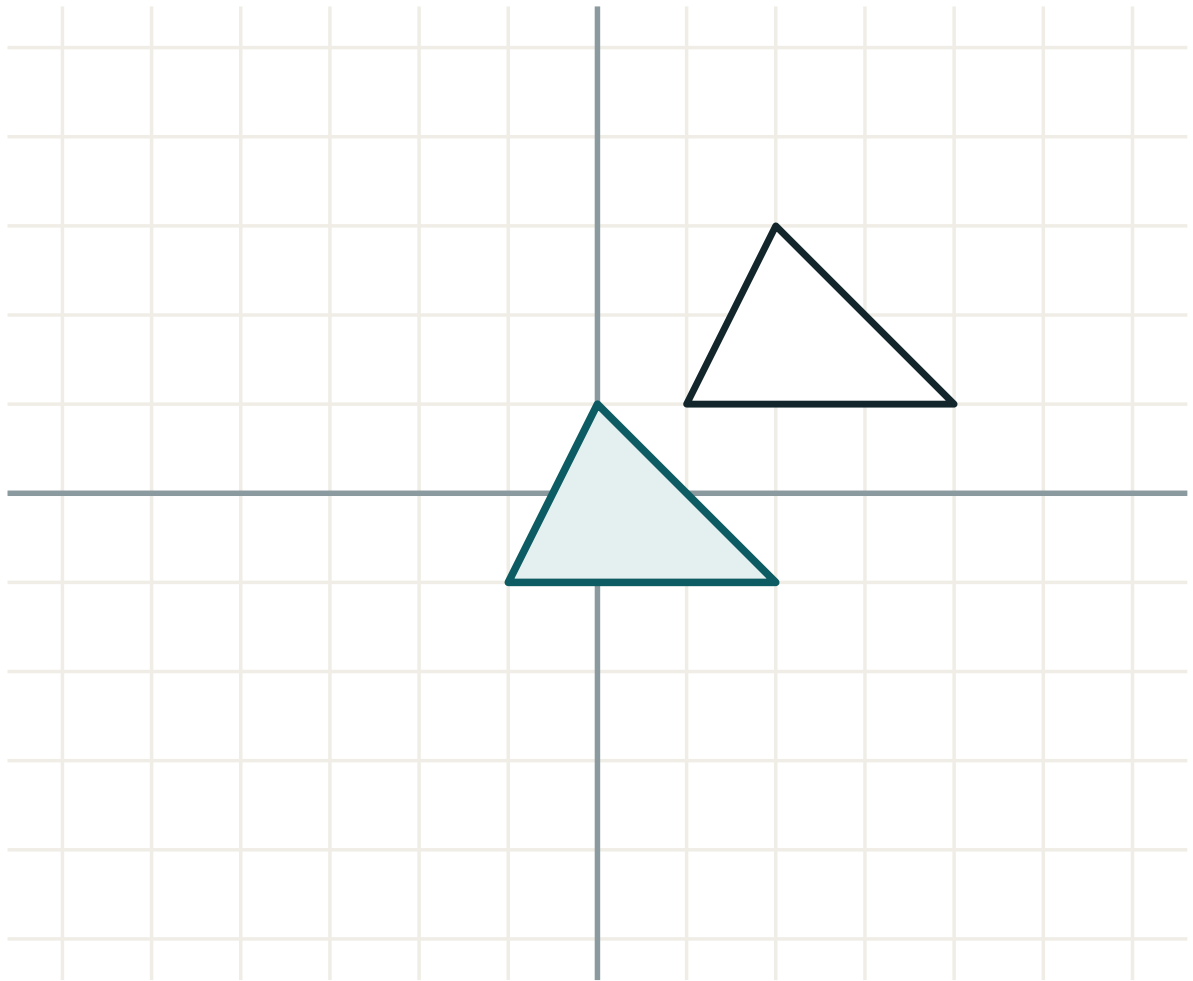
ANSWER

$a \parallel c$

03 Interactive model

Ask the class to draw two lines that “look parallel” and then measure the gap at both ends of the page; the ones that fail are the lesson.

A translated line is parallel to the original

transformation **translation**lengths **unchanged**angles **unchanged**orientation **kept**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Parallel lines	Parallel to'g'ri chiziqlar	Параллельные прямые
To intersect	Kesishmoq	Пересекаться
Plane	Tekislik	Плоскость
Axiom	Aksioma	Аксиома
Distance	Masofa	Расстояние
Direction	Yo'nalish	Направление
Unique	Yagona	Единственный
Symbol	Belgi	Обозначение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Parallel lines have how many common points?

The symbol for parallel is:

Through a point off a line, parallels number:

The distance between parallel lines is:

increasing

constant

decreasing

zero

Distance from a point to a line is measured:

any way

along the perpendicular

along the line

diagonally

If $a \parallel b$ and $b \parallel c$ then:

$a \perp c$

$a \parallel c$

$a = c$

nothing follows

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Two lines with no common point are

ANSWER parallel

2. Write “ a is parallel to b ” in symbols

ANSWER $a \parallel b$

3. Two lines crossing at one point are

ANSWER intersecting

4. Parallels through a point off a line

ANSWER 1

5. The distance between parallels is

ANSWER constant

6. Distance to a line is measured along the

ANSWER perpendicular

7. Name one parallel pair in the classroom

ANSWER e.g. the opposite edges of the board

Medium · the standard question

7 PROBLEMS

1. Gaps of 3 and 3.4 *cm*: parallel?

ANSWER No

2. $a \parallel b, b \parallel c$: is $a \parallel c$?

ANSWER Yes

3. Can two parallel lines lie in different planes?

ANSWER Not by this definition — the same plane is required

4. Is “exactly one parallel” a theorem or an axiom?

ANSWER An axiom

5. Is “at least one parallel” a theorem or an axiom?

ANSWER A theorem

6. Two rails 1520 *mm* apart everywhere are

ANSWER parallel

7. Why is looking at a drawing not a proof?

ANSWER The lines may meet beyond the page

Hard · stretch and reason

7 PROBLEMS

1. Prove that $a \parallel b$ and $b \parallel c$ force $a \parallel c$

ANSWER Two parallels to b through one point contradict the axiom

2. If $a \perp c$ and $b \perp c$ in a plane, then

ANSWER $a \parallel b$

3. How many lines are parallel to a in the whole plane?

ANSWER Infinitely many — one through each point off it

4. A quadrilateral with both pairs of opposite sides parallel is a

ANSWER parallelogram

5. Explain why the axiom cannot be checked by measurement

ANSWER Measurement covers a finite part of an endless line

6. What fails without the axiom of parallels?

ANSWER The angle sum of a triangle is no longer 180°

7. Two lines are 0 apart at one point: what are they?

ANSWER Intersecting there, or the same line

07 Homework

Homework — 5 tasks

Measure, do not eyeball: take the gap at both ends of the page.

1. Draw a line a and a point M off it, and construct the parallel through M .
2. Find five parallel pairs at home and say what each is for.
3. State the axiom of parallels in your own words.
4. Two lines are 4 cm apart at the left edge and 4 cm apart at the right. Are they parallel? What else would you check?

5. Explain why $a \parallel b$ and $b \parallel c$ give $a \parallel c$.